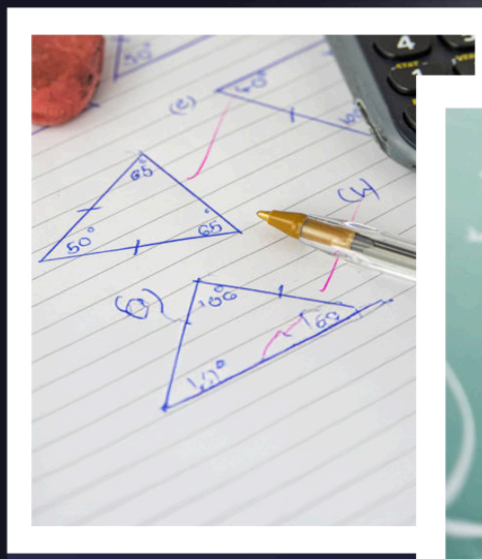
testbook PASS PRO MAX

# Quantitative Aptitude

## Formula – Guide

## Average



# Average

## Average

- Average formula = $\frac{Some\ of\ all\ the\ observations}{Total\ no.\ of\ observations}$

  = $(X_1, X_2, X_3, \ldots\ldots, X_n) / n$

## Combined average

- If $x_1$ and $x_2$, respectively represent the means of the $n_1$ and $n_2$ observations, then the combined average or combined mean is

  Combined average = $\frac{(n_1x_1+n_2x_2)}{n_1+n_2}$

## Different problems on average

- **Batting average =** $\frac{Total\ runs\ scored}{Total\ number\ of\ innings\ played}$
- **Bowling average =** $\frac{Total\ runs\ given}{Total\ wickets\ taken}$
- **Average of consecutive numbers =** $\frac{First\ number + last\ number}{2}$
- **Average speed =** $\frac{Total\ distance}{Total\ time\ taken}$

## For consecutive natural numbers

- Average of first n natural numbers = $\frac{(n+1)}{2}$
- Average of Square of first n natural numbers = $\frac{n(n+1)(2n+1)}{6}$
- Average of cube of first n natural numbers = $\frac{n(n+1)^2}{4}$

## For Consecutive Even Numbers

- Average of first n even numbers = (n+1)
- Average of Square of first n even numbers = $\frac{2(n+1)(2n+1)}{3}$
- Average of cube of first n even numbers = $2n(n+1)^2$

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## For Consecutive odd Numbers

- Average of first n even numbers = n
- Average of Square of first n even numbers = $\frac{(2n+1)(2n-1)}{3}$
- Average of cube of first n even numbers = $n(2n^2-1)$

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## Important results

- If average of n observations is a but the average becomes b when one observation is eliminated, then value of eliminated observation = **n (a – b) + b**
- Average of n observations is a but the average becomes b when a new observation is added, then value of added observation = **n (b–a) + b**
- We have n observations out of which some observations ($a_1$, $a_2$ , $a_3$.....) are replaced by some other new observations and in this way, if the average increase or decreases by b, then Value of new observations = **a±nb**

  (In this formula, the signs of '+' and '–' depend upon the increment or decrement in the average)
- If the average of n students in a class is a, where average of passed students is x and average of failed students is y, then

  Number of students passed = $\frac{n(a-y)}{x-y}$

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